American Computer Science League

2022-2023 ● Contest 3: Short Problems ● Elementary Division

1. Boolean Algebra

Of the whole numbers 1 through 13 inclusive, how many satisfy this Boolean expression?

$$(x \geq 6)\ AND\ (x < 8)\ OR\ NOT\ (x < 10)$$

A. 2
B. 5
C. 6
D. 7
E. 10

2. Boolean Algebra

Determine how many ordered pair(s) make the following expression TRUE:

$$\sim(\sim A \sim B)\ +\ \sim(A\ +\ B)$$

A. 0
B. 1
C. 2
D. 3
E. 4

3. Boolean Algebra

Determine which of the following are equivalent Boolean expressions:

a. $\sim B(A \sim B\ +\ A)$ b. $\sim A(B\ +\ A)$ c. $\sim(A\ +\ B)$ d. $\sim A \sim B$

A. a,b,c
B. b,c,d
C. a,d
D. c,d
E. b,c

4. Boolean Algebra

Simplify the following Boolean expression:

$$\sim AB(\sim A\ +\ \sim B)\ +\ A \sim B(\sim A\ +\ \sim B)$$

A. $AB\ +\ A \sim B$
B. $A \sim B\ +\ \sim AB$
C. $AB\ +\ \sim A \sim B$
D. $A\ +\ \sim B$
E. $\sim A\ +\ B$

5. Boolean Algebra

Determine which of the following is NOT a tautology:

a. $A(\sim B\ +\ B)\ +\ \sim A$ b. $\sim(A\ +\ B)\ +\ (A\ +\ B)$
c. $A(A \sim B)\ +\ B(\sim AB)$ d. $AB\ +\ \sim AB\ +\ \sim B$

A. a
B. b
C. c
D. d
E. They all are.

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1. 布尔代数

从 1 到 13（**包含 1 和 13**）**中**有多少个整数满足**以下**布尔表达式？

$$(x \geq 6)\ AND\ (x < 8)\ OR\ NOT\ (x < 10)$$

A. 2
B. 5
C. 6
D. 7
E. 10

2. 布尔代数

有多少个有序对使得**以下**表达式**的**结果为真？

$$\sim(\sim A \sim B)\ +\ \sim(A\ +\ B)$$

A. 0
B. 1
C. 2
D. 3
E. 4

3. 布尔代数

以下哪些布尔表达式等价？

a. $\sim B(A \sim B\ +\ A)$ b. $\sim A(B\ +\ A)$ c. $\sim(A\ +\ B)$ d. $\sim A \sim B$

A. a,b,c
B. b,c,d
C. a,d
D. c,d
E. b,c

4. 布尔代数

化简**以下布尔表达式：**

$$\sim AB(\sim A\ +\ \sim B)\ +\ A \sim B(\sim A\ +\ \sim B)$$

A. $AB\ +\ A \sim B$
B. $A \sim B\ +\ \sim AB$
C. $AB\ +\ \sim A \sim B$
D. $A\ +\ \sim B$
E. $\sim A\ +\ B$

5. 布尔代数

以下哪项**不是重言式**？

a. $A(\sim B\ +\ B)\ +\ \sim A$ b. $\sim(A\ +\ B)\ +\ (A\ +\ B)$
c. $A(A \sim B)\ +\ B(\sim AB)$ d. $AB\ +\ \sim AB\ +\ \sim B$

A. a
B. b
C. c
D. d
E. 以上**全部**